

Course Code : ECT 359-4

MPNO/MS – 24 / 1670

Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination

INTRODUCTION TO INTERNET OF THINGS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
 - (2) Due credit will be given to neatness.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1. (a) For a Self Driven Cars Use Case, recommend an IoT architecture with details and justification. 7(CO1,5)
 (b) As an IoT Application Developer, select an Access Technology for Swarm Drone applications. Justify. 4(CO1,5)

 2. (a) Compare Raspberry pi with cameras setup and OpenMV- H7 for a Industrial Robot application. 4(CO1)
 (b) Describe the steps to execute a python script after loading the desktop environment on RPi. 3(CO2)

 3. (a) Write a bash script to execute a script "test2.sh" in a loop with a two input argument stored in arrays of length 16 and 4, respectively. 6(CO2)
 (b) Write a bash script to clone, compile, build and install source code of RPi-GPIO python library located at a git repository. 4(CO1,3,4)

 4. (a) Write a python script to perform moving average of five elements of an array X = [1 2 3 4 5 6 7 8 9]. 5(CO1,3,4)

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Contd.

- (b) Write a python script to read an analog temperature sensor value deployed on a manufacturing plant in a remote area using pi to control a cooling fan locally driven by a relay, Fan should be ON above 45⁰C and otherwise OFF. 7(CO1,3,4)
5. (a) Write a python script to interface ADC with pi and acquire an analog signal from Hall effect sensor to control the speed of the DC motor. 6(CO1,3,4)
- (b) Compare I2C and SPI protocol with examples. 4(CO4)
6. (a) Identify the challenges of implementing an IoT Use case for Robotic Surgery in India. How would you overcome those ? Justify. 6(CO1,5)
- (b) Explain the MQTT protocol with example. 4(CO1)

